Week 07 teacher guide

Area and surface area

PowerPoint slides 577-672 | Student workbook pp. 54-61

Area counts square units that cover a surface. Cutting and rearranging reveal triangle and parallelogram formulas. A net helps count every face of a solid.

The seven days

Day 043 · p. 54 · Saturday · 20 questions,including 2 challenges
Area from a base and height

Day 044 · pp. 55-56 · Sunday · 20 questions,including 2 challenges
Nets and surface area

Day 045 · p. 57 · Monday · 20 questions,including 2 challenges
Area and surface area

Day 046 · p. 58 · Tuesday · 20 questions,including 2 challenges
Area and surface area

Day 047 · p. 59 · Wednesday · 25 questions,including 2 challenges
Area and surface area

Day 048 · p. 60 · Thursday · 20 questions,including 2 challenges
Area and surface area

Day 049 · p. 61 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

The student pages contain questions only. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons use 20 core questions including C1 and C2. Weekdays use 20 questions; Wednesday uses 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The Sunday reference page is for checking a method after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer-review sections are optional after Friday and do not add to the 80-minute weekend lesson.

Built-in review

Day 045: Area, surface area and volume (9, 10, 11, 12)

Day 046: Area, surface area and volume (9, 10, 11, 12)

Day 047: Weekly puzzle (13)

Day 048: Expressions and equations (9, 10, 11, 12)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Saturday. Area from a base and height

Day 043 · p. 54 | Slides 577-606

Goal: Find area by matching a base with its perpendicular height.

Check first: Rectangle area and multiplication with fractions.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 577-578 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 579-580 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 581-582 · Explore · P1, P2

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 583 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 584-587 · Explore · P3, P4, P5, P6

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 588-590 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 591 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 592 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 593-596 · Guided practice · X1, X2, X3, X4

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 597-598 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 599-602 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 603 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 604 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 605 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 606 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: Slide the vertex, keep the area

Watch and discuss; this is a projected investigation, not an extra assigned question.

Triangles ABC and ABD share A(0, 0) and B(8, 0). Their third vertices are C(2, 6) and D(10, 6). Predict whether their areas match. Find both areas, draw both perpendicular heights, and explain why the second height falls outside its triangle.

1. Locate the common base on y = 0: its length is 8.
2. Both top vertices are 6 units above that line, so both matching heights are 6.
3. Each area is one half of 8 × 6 = 24; the slanted sides need not match.

Ask: Which distance stays fixed while the vertex slides?

If stuck: Extend the base line before dropping a perpendicular from D.

Expected: Both areas 24 square units; both matching heights 6 units.

C1 · Challenge discussion

Ask: Which two lengths measure perpendicular distances?

If stuck: Draw the height as a right-angle segment, then subtract the pond.

Expected: 48 square m.

C2 · Challenge discussion

Ask: What does area measure that perimeter does not?

If stuck: Find the common area before solving the missing rectangle side.

Expected: Rectangle 5 × 12 cm; both areas 60 square cm. Perimeters 34 cm and 36 cm; not equal.

What to notice

Marks a right angle at the foot of the height. → Ask whether the slanted side makes that angle.

Rearranges without gaps or overlaps. → Compare the original and rearranged boundaries.

Common error: Uses a slanted side as height. Repair: Trace the perpendicular distance with a ruler.

Sunday. Nets and surface area

Day 044 · pp. 55-56 | Slides 607-636

Goal: Use decompositions and nets to count every part of a surface.

Check first: Rectangle and triangle area; multiplication and addition.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 607-608 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 609-610 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 611 · Compare methods · P1

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 612 · Explore · P2

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 613 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 614-616 · Explore · P3, P4, P5

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 617-619 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 620 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 621 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 622 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 623 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 624 · Explore · M1, M2, M3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 625-626 · Guided practice · X1, X2

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 627-628 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 629-632 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 633 · Summary · S

Ask for one sentence or annotated example in the learner's own words. Listen for the main relationship.

Slides 634 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 635 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 636 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Sunday. Discussion and support

Investigation T: Audit an open-box net

Watch and discuss; this is a projected investigation, not an extra assigned question.

An open box has length 6, width 4, and height 2 cm. A learner adds 24 + 8 + 8 = 40 square cm for its material. Identify the missing faces with a net. Repair the total, then compare with the corresponding closed box.

1. Name the bottom: 6 × 4 = 24.
2. Name two 6 × 2 walls and two 4 × 2 walls: 24 + 16.
3. Add 64 for the open box; add the 24-square-centimeter lid to get 88 closed.

Ask: Which physical face does each number in the learner'ssum represent?

If stuck: Fold a rough net from plain paper and name the bottom and four walls.

Expected: Two 6-by 2 side faces are missing. Open box 64 square cm; closed box 88 square cm.

C1 · Challenge discussion

Ask: Which face is removed, and which piece is added?

If stuck: List the six face areas before changing the design.

Expected: Closed 52 square cm; second 46 square cm; 6 square cm saved.

C2 · Challenge discussion

Ask: Which faces change when only length changes?

If stuck: Keep the two triangular ends separate from the three rectangles.

Expected: At 10 cm: surface area 132 square cm, volume 60 cubic cm. At 20 cm: surface area 252 square cm, volume 120 cubic cm. Only volume doubles.

What to notice

Labels both hidden and visible faces. → Match each label on the solid and the net.

Uses different decompositions for the L. → Accept both and compare the totals.

Common error: Uses length × width × height for surface area. Repair: Imagine covering every face with paper.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Saturday

577. Week 7, Saturday
Watch and discuss

578. Today
Watch and discuss

579. Recall
Day 043 · Workbook p. 54 · R1, R2, R3, R4

580. Recall: answers
Day 043 · Workbook p. 54 · R1, R2, R3, R4

581. Rectangle area review
Day 043 · Workbook p. 54 · P1

582. Parallelogram
Day 043 · Workbook p. 54 · P2

583. The model
Watch and discuss

584. Right triangle
Day 043 · Workbook p. 54 · P3

585. Height is not a side
Day 043 · Workbook p. 54 · P4

586. Obtuse triangle
Day 043 · Workbook p. 54 · P5

587. Rational dimensions
Day 043 · Workbook p. 54 · P6

588. Worked example · 1 of 3
Watch and discuss · worked example

589. Worked example · 2 of 3
Watch and discuss · worked example

590. Worked example · 3 of 3
Watch and discuss · worked example

591. Your turn
Day 043 · Workbook p. 54 · F

592. A claim to check
Day 043 · Workbook p. 54 · D

593. Guided practice · X1
Day 043 · Workbook p. 54 · X1

594. Guided practice · X2
Day 043 · Workbook p. 54 · X2

595. Guided practice · X3
Day 043 · Workbook p. 54 · X3

596. Guided practice · X4
Day 043 · Workbook p. 54 · X4

597. Slide the vertex, keep the area
Watch and discuss · draw on separate paper

598. Slide the vertex, keep the area
Watch and discuss · draw on separate paper

599. Challenge C1
Day 043 · Workbook p. 54 · C1

600. Challenge C1 · Discuss
Day 043 · Workbook p. 54 · C1

601. Challenge C2
Day 043 · Workbook p. 54 · C2

602. Challenge C2 · Discuss
Day 043 · Workbook p. 54 · C2

603. In your own words
Day 043 · Workbook p. 54 · S

604. Exit ticket
Day 043 · Workbook p. 54 · E

605. The week ahead
Days 045-049 · Workbook pp. 57-61

606. Tutor reference
Tutor only;hidden in slide show

Sunday

607. Week 7, Sunday
Watch and discuss

608. Today
Watch and discuss

609. Recall
Day 044 · Workbook p. 55 · R1, R2, R3, R4

610. Recall: answers
Day 044 · Workbook p. 55 · R1, R2, R3, R4

611. L-shaped polygon split two ways
Day 044 · Workbook p. 55 · P1

612. Rectangular prism net
Day 044 · Workbook p. 55 · P2

613. The model
Watch and discuss

614. Triangular prism net
Day 044 · Workbook p. 55 · P3

615. Square pyramid net
Day 044 · Workbook p. 55 · P4

616. Find a missing face in a surface-area calculation
Day 044 · Workbook p. 55 · P5

617. Worked example · 1 of 3
Watch and discuss · worked example

618. Worked example · 2 of 3
Watch and discuss · worked example

619. Worked example · 3 of 3
Watch and discuss · worked example

620. Your turn
Day 044 · Workbook p. 55 · F

621. Two ways to see it
Watch and discuss · compare the same example

622. A claim to check
Day 044 · Workbook p. 55 · D

623. Words and meaning
Watch and discuss

624. Mixed review
Day 044 · Workbook p. 55 · M1, M2, M3

625. Guided practice · X1
Day 044 · Workbook p. 55 · X1

626. Guided practice · X2
Day 044 · Workbook p. 55 · X2

627. Audit an open-box net
Watch and discuss · draw on separate paper

628. Audit an open-box net
Watch and discuss · draw on separate paper

629. Challenge C1
Day 044 · Workbook p. 55 · C1

630. Challenge C1 · Discuss
Day 044 · Workbook p. 55 · C1

631. Challenge C2
Day 044 · Workbook p. 55 · C2

632. Challenge C2 · Discuss
Day 044 · Workbook p. 55 · C2

633. In your own words
Day 044 · Workbook p. 55 · S

634. Exit ticket
Day 044 · Workbook p. 55 · E

635. The week ahead
Days 045-049 · Workbook pp. 57-61

636. Tutor reference
Tutor only;hidden in slide show

Answer review

637. Week 7 answers
Next lesson: slide 673. Use this review section after Friday.

638. Area from a base and height · 1 of 5
Day 043 · Workbook p. 54 · Items R1, R2, R3, R4, P1

639. Area from a base and height · 2 of 5
Day 043 · Workbook p. 54 · Items P2, P3, P4, P5, P6

640. Area from a base and height · 3 of 5
Day 043 · Workbook p. 54 · Items F, D, X1, X2, X3

641. Area from a base and height · 4 of 5
Day 043 · Workbook p. 54 · Items X4, S, E, C1

642. Area from a base and height · 5 of 5
Day 043 · Workbook p. 54 · Items C2

643. Nets and surface area · 1 of 5
Day 044 · Workbook p. 55 · Items R1, R2, R3, R4, P1

644. Nets and surface area · 2 of 5
Day 044 · Workbook p. 55 · Items P2, P3, P4, P5

645. Nets and surface area · 3 of 5
Day 044 · Workbook p. 55 · Items F, D, M1, M2, M3

646. Nets and surface area · 4 of 5
Day 044 · Workbook p. 55 · Items X1, X2, S, E

647. Nets and surface area · 5 of 5
Day 044 · Workbook p. 55 · Items C1, C2

648. Area and surface area · 1 of 4
Day 045 · Workbook p. 57 · Items 1, 2, 3, 4, 5

649. Area and surface area · 2 of 4
Day 045 · Workbook p. 57 · Items 6, 7, 8, 9, 10

650. Area and surface area · 3 of 4
Day 045 · Workbook p. 57 · Items 11, 12, 13, 14, 15

651. Area and surface area · 4 of 4
Day 045 · Workbook p. 57 · Items 16, 17, 18, C1, C2

652. Area and surface area · 1 of 5
Day 046 · Workbook p. 58 · Items 1, 2, 3, 4, 5

653. Area and surface area · 2 of 5
Day 046 · Workbook p. 58 · Items 6, 7, 8, 9, 10

654. Area and surface area · 3 of 5
Day 046 · Workbook p. 58 · Items 11, 12, 13, 14, 15

655. Area and surface area · 4 of 5
Day 046 · Workbook p. 58 · Items 16, 17, 18, C1

656. Area and surface area · 5 of 5
Day 046 · Workbook p. 58 · Items C2

657. Area and surface area · 1 of 6
Day 047 · Workbook p. 59 · Items 1, 2, 3, 4, 5

658. Area and surface area · 2 of 6
Day 047 · Workbook p. 59 · Items 6, 7, 8, 9

659. Area and surface area · 3 of 6
Day 047 · Workbook p. 59 · Items 10, 11, 12, 13, 14

660. Area and surface area · 4 of 6
Day 047 · Workbook p. 59 · Items 15, 16, 17, 18, 19

661. Area and surface area · 5 of 6
Day 047 · Workbook p. 59 · Items 20, 21, 22, 23, C1

662. Area and surface area · 6 of 6
Day 047 · Workbook p. 59 · Items C2

663. Area and surface area · 1 of 5
Day 048 · Workbook p. 60 · Items 1, 2, 3, 4, 5

664. Area and surface area · 2 of 5
Day 048 · Workbook p. 60 · Items 6, 7, 8, 9, 10

665. Area and surface area · 3 of 5
Day 048 · Workbook p. 60 · Items 11, 12, 13, 14, 15

666. Area and surface area · 4 of 5
Day 048 · Workbook p. 60 · Items 16, 17, 18, C1

667. Area and surface area · 5 of 5
Day 048 · Workbook p. 60 · Items C2

668. Weekly check · 1 of 5
Day 049 · Workbook p. 61 · Items 1, 2, 3, 4, 5

669. Weekly check · 2 of 5
Day 049 · Workbook p. 61 · Items 6, 7, 8, 9

670. Weekly check · 3 of 5
Day 049 · Workbook p. 61 · Items 10, 11, 12, 13, 14

671. Weekly check · 4 of 5
Day 049 · Workbook p. 61 · Items 15, 16, 17, 18

672. Weekly check · 5 of 5
Day 049 · Workbook p. 61 · Items C1, C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 043 · Area from a base and height

Workbook p. 54 · assigned set

R1 · F-DIV

$3\frac{1}{3}$

R2 · D-MULT

$7.35

R3 · E-EXP

9

R4 · F-DIV

6

P1 · G-AREA

Area 15 square cm; perimeter 17 cm

P2 · G-PARA

28 square cm

P3 · G-TRI

24 square cm both ways

P4 · G-TRI

60 square cm

P5 · G-TRI

6 square cm

P6 · G-TRI

6 square cm; no, the height must be perpendicular

F · G-TRI

2.5; 10 square cm

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · G-TRI

The triangle is half the parallelogram: 18 square cm.

Reason: Counts both copies of the triangle. Place two matching paper triangles together.

X1 · G-AREA

18 square cm; 18 cm.

X2 · G-PARA

12 square cm; slanted side 5 cm unused.

X3 · G-TRI

12 square cm.

X4 · G-TRI

3 cm.

S · EXPLAIN

Answers vary. A parallelogram rearranges into a rectangle. Two copies of a triangle make a parallelogram.

Reason: Accept an accurate explanation supported by a matching example.

E · G-TRI

48 square cm; height 8 cm

Reason: Half of the 12 by 8 surrounding rectangle is 48.

C1 · G-PARA

48 square m.

Reason: Garden area = 12 × 5 = 60. Pond area = 6 × 4 ÷ 2 = 12. Planted area = 60 − 12 = 48. The 7 m side is slanted; it does not measure perpendicular distance between the parallel bases.

C2 · G-TRI

Rectangle 5 × 12 cm; both areas 60 square cm. Perimeters 34 cm and 36 cm; not equal.

Reason: Triangle area = 10 × 12 ÷ 2 = 60. A rectangle with side 5 needs the other side 60 ÷ 5 = 12. Its perimeter is 34; the given triangle perimeter is 36. Equal area does not force equal boundary length.

Day 044 · Nets and surface area

Workbook p. 55 · assigned set

R1 · F-DIV

$2\frac{1}{2}$; stories vary

R2 · D-SUB

7.714

R3 · E-ADD

x = 14

R4 · F-DIV

1 meters

P1 · G-COMP

42 square cm

P2 · G-SA

Surface area 52 square cm; volume 24 cubic cm

P3 · G-SA

Two triangles of area 24 each; rectangles of area 30, 40, 50; total 168 square cm

P4 · G-SA

96 square cm

P5 · G-SA

76 square cm

F · G-SA

3; 40 square cm

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · G-SA

A closed cube has six faces: 6 × 9 = 54 square cm.

Reason: One hidden face was omitted. Label each face of the folded cube and net with matching letters.

M1 · F-DIV

12 pieces

M2 · N-DIV

13

M3 · E-RELATE

(1, 2), (3, 6), (5, 10)

X1 · G-SA

76 square cm.

X2 · G-SA

96 square cm; 64 cubic cm.

S · EXPLAIN

Answers vary. A net unfolds every face. Surface area adds face areas in square units.

Reason: Accept an accurate explanation supported by a matching example.

E · G-SA

108 square cm

Reason: Two triangle ends total 12. Three rectangles total 24 + 32 + 40 = 96. Total 108.

C1 · G-SA

Closed 52 square cm; second 46 square cm; 6 square cm saved.

Reason: The closed box needs 2(12 + 8 + 6) = 52. Remove the 12-square-cm top and add the 6-square-cm label: 52 − 12 + 6 = 46. Saving = 52 − 46 = 6.

C2 · G-SA

At 10 cm: surface area 132 square cm, volume 60 cubic cm. At 20 cm: surface area 252 square cm, volume 120 cubic cm. Only volume doubles.

Reason: Each end area is 6. The side rectangles total (3 + 4 + 5) × length. SA = 12 + 12l, so it changes 132→252. Volume = 6l, changing 60→120. The two ends do not grow when only the prism length changes.

Day 045 · Area and surface area

Workbook p. 57 · assigned set

1 · N-FACT

27

2 · N-FACT

63

3 · N-FACT

36

4 · N-FACT

81

5 · N-DIV

42

6 · P-PERCENT

75 %

7 · N-FACT

15

8 · G-PARA

48 square cm

9 · G-AREA · review

9 square cm

10 · G-PARA · review

36 square cm

11 · G-TRI · review

18 square cm

12 · G-COMP · review

39 square cm

13 · G-TRI

The triangle is half the parallelogram: 18 square cm.

14 · G-AREA

40 square cm; 26 cm.

15 · G-PARA

30 square cm; slanted side 7 cm unused.

16 · G-TRI
30 square cm.

17 · G-TRI
5 cm.

18 · G-COMP
57 square cm.

C1 · G-COMP
20 square m to paint; 4 liters.

Reason: Whole wall 24; window 3; vent 1. Paint area = 24 − 3 − 1 = 20. Divide by 5 to get 4 liters. Only the uncovered wall is painted.

C2 · G-TRI
Heights 6 cm and 8 cm; B is $\frac{4}{3}$ times as tall.

Reason: For fixed area 24, base × height = 48. Divide 48 by each base: 6 and 8. The height factor is $8 \div 6 = \frac{4}{3}$. A smaller base needs a greater height to preserve area.

Day 046 · Area and surface area

Workbook p. 58 · assigned set

1 · F-EQUIV
0.5; 50%

2 · F-EQUIV
0.25; 25%

3 · F-EQUIV
0.75; 75%

4 · F-EQUIV
0.125; 12.5%

5 · E-EXP
9

6 · F-DIV
$2\frac{1}{2}$; stories vary

7 · N-FACT
12

8 · G-SA
66 square cm

9 · G-SA · review
62 square cm

10 · G-SA · review
60 square cm

11 · G-SA · review
40 square cm

12 · G-SA · review
24 square cm

13 · G-SA
A closed cube has six faces: 6 × 9 = 54 square cm.

14 · G-AREA
54 square cm; 30 cm.

15 · G-PARA
42 square cm; slanted side 8 cm unused.

16 · G-TRI
42 square cm.

17 · G-TRI
6 cm.

18 · G-COMP
74 square cm.

C1 · G-SA
Both 36 cubic cm; surface areas 72 and 66 square cm. The 3 × 3 × 4 box saves 6 square cm.

Reason: Volumes are 2 × 3 × 6 and 3 × 3 × 4, both 36. Surface areas are 2(6 + 12 + 18) = 72 and 2(9 + 12 + 12) = 66. Equal volume does not imply equal surface area.

C2 · G-SA
$12 for four faces; $9.60 extra for the base; total $21.60.

Reason: One face area = 8 × 5 ÷ 2 = 20; four total 80. At $0.15 each square cm, cost = $12. Base area 64 costs 64 × 0.15 = $9.60 extra.

Day 047 · Area and surface area

Workbook p. 59 · assigned set

1 · N-SCALE
92

2 · N-SCALE
9

3 · N-SCALE
19

4 · N-SCALE
27

5 · G-COMP
38 square cm

6 · G-PARA
24 square cm

7 · E-ADD

x = 14

8 · D-MULT

0.56; yes

9 · F-MULT

$\frac{3}{5}$

10 · F-COMP

$\frac{2}{5}$, 0.45, $\frac{1}{2}$

11 · N-FACT

10

12 · G-VOL

30 cm³

13 · G-AREA · review

5 cm and 7 cm

14 · G-AREA

30 square cm; 26 cm.

15 · G-PARA

24 square cm; slanted side 5 cm unused.

16 · G-TRI

24 square cm.

17 · G-TRI

3 cm.

18 · G-COMP

49 square cm.

19 · G-TRI

Triangle 12; parallelogram 24; ratio 1: 2.

20 · G-SA

92 square cm.

21 · G-SA

54 square cm; 27 cubic cm.

22 · G-SA

132 square cm.

23 · G-SA

108 square cm.

C1 · G-COMP

6 packs; unused tiles cover 4 square m.

Reason: Floor area = 80 − 12 = 68, requiring 34 tiles. Five packs give only 30 tiles; six give 36. Two unused tiles cover 4 square m.

C2 · G-SA

84 square m of fabric; closed 108 square m; floor adds 24.

Reason: Each triangle has area 3 × 4 ÷ 2 = 6, so ends total 12. The other side rectangles are 4 × 8 and 5 × 8, totaling 72. Fabric = 84. The missing 3 × 8 floor adds 24, giving 108 closed.

Day 048 · Area and surface area

Workbook p. 60 · assigned set

1 · N-EST

900

2 · N-EST

10

3 · N-EST

About 100

4 · N-EST

About 20

5 · G-COMP

38 square cm

6 · G-PARA

24 square cm

7 · G-SA

62 square cm

8 · E-EXPR

8b; 32 beads

9 · E-DIST · review

3x + 18

10 · E-EXP · review

16

11 · E-MULT · review

x = 9

12 · E-RELATE · review

(0, 0), (2, 8), (3, 12)

13 · G-SA

120 square cm

14 · G-SA

A closed cube has six faces: 6 × 9 = 54 square cm.

15 · G-AREA

44 square cm; 30 cm.

16 · G-PARA

36 square cm; slanted side 6 cm unused.

17 · G-TRI

36 square cm.

18 · G-TRI

4 cm.

C1 · G-COMP

153 square cm.

Reason: One remaining sign has area 63 − 12 = 51. Three total 3 × 51 = 153. Equivalently, start with 3 × 63 and subtract 3 × 12. Distributing 3 preserves the same result.

C2 · G-SA

64 and 96 square cm; difference 32; open box 80 square cm, so neither total.

Reason: Each face is 16. Four faces give 64; six give 96. Removing one top from the closed cube leaves 5 × 16 = 80. The side-only count also omits the bottom.

Day 049 · Weekly check

Workbook p. 61 · assigned set

1 · G-COMP

38 square cm

2 · G-PARA

24 square cm

3 · G-SA

62 square cm

4 · G-SA

96 square cm

5 · E-EVAL

11

6 · D-DIV

0.375

7 · E-MULT

x = 3.2

8 · F-MULT

$\frac{3}{5}$

9 · F-COMP

$\frac{2}{5}$, 0.45, $\frac{1}{2}$

10 · G-TRI

The triangle is half the parallelogram: 18 square cm.

11 · G-AREA

60 square cm; 34 cm.

12 · G-PARA

50 square cm; slanted side 7 cm unused.

13 · G-TRI

50 square cm.

14 · G-TRI

5 cm.

15 · G-COMP

85 square cm.

16 · G-TRI

Triangle 25; parallelogram 50; ratio 1: 2.

17 · G-SA

160 square cm.

18 · G-SA

150 square cm; 125 cubic cm.

C1 · G-TRI

Parallelogram height 8 cm; triangle height 16 cm; reduced-height triangle area 58.5 square cm.

Reason: Parallelogram height = 72 ÷ 9 = 8. Triangle needs base × height = 144, so height 16. Halving a base-height product explains the factor 2. New height 13 gives 9 × 13 ÷ 2 = 58.5.

C2 · G-SA

128 square cm; $2.56.

Reason: Surface area = 2(20 + 24 + 30) = 148. Subtract the 20 already covered to get 128. Cost = 128 × 0.02 = $2.56. Wrapping covers faces, not interior space.